Joanne Hoar

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Senior Software Developer

WHO AM I?

Canadian Software Developer with 20 years of experience in video game programming, scientific modelling and cross-platform development. Expertise in multiple programming languages, game engines and SDKs. Proven track record in leading teams, mentoring developers, exceeding client expectations and building corporate culture. Strong background in C++, Unreal Engine, iOS and Android using agile methodologies and CI/CD. Passion for algorithmic correctness, mathematics, artificial intelligence and nature.

EXPERIENCE

Jan 2025 – Present **Technology Instructor** **Robertson College (contract)**

- Instructed 20 course offerings across 12 subjects for Full Stack Development (FSD) and Data Analytics (DA) diploma programs.
- Courses taught: C# Programming, .NET, ASP .NET, ASP .NET Core, Object-oriented Programming (Java), Front End Frameworks, Intro to JavaScript, Web Development Basics, Data Collection, Testing and Quality Assurance, Version Control with Git and GitHub, Introduction to IT.
- Deliver a minimum of 3 synchronous live sessions per week per course, hosting live coding sessions and debugging student projects in real-time.
- Manage and maintain a GitHub repository hosting open-source course materials and sample code.
- Archive session recordings for asynchronous review and participate in ongoing monthly instructor training seminars.
- Maintained an average student satisfaction rating of 4.5/5.0 across evaluated courses, with feedback highlighting approachability, real-world teaching approach, and responsiveness.

`C# / .NET / ASP .NET Core / Java / JavaScript / React / Front End Frameworks / Git / GitHub / Azure / SQL / Testing and QA / Data Collection`

2022 – 2025 **Senior Software Developer** **bitHeads Inc.**

- Developed and maintained cross-platform client SDKs for brainCloud BaaS, enabling gamification, leaderboards, matchmaking, lobbies, real-time multiplayer, and cloud code for 50+ published games across 12 platforms including PlayStation, Xbox, Nintendo Switch, Steam, iOS, and Android.
- Primary focus on C++ client SDK, Unreal Engine plugin, iOS (Objective-C/Swift), Android Native (Java/Kotlin), and PlayStation; reviewed and debugged all other supported platforms and languages.
- Built and maintained Unreal Engine sample projects including Space Shooter, Target Practice, Tappy Chicken, Relay Blast, and dedicated server demo to help onboard client studios.
- Supported integration for notable titles including SUPERHOT VR (SUPERHOT Team), Tony Hawk's Skate Jam (Maple Media), Crypt of the NecroDancer (Brace Yourself Games), and Invizimals (PlayStation Mobile).
- Engineered and maintained CI/CD and TDD pipelines ensuring high-quality multi-platform releases.
- Designed and built demonstration applications and accompanying widget libraries for every supported client platform to showcase SDK capabilities — UI frameworks spanned Dear ImGui (C++), Unreal Engine UMG, SwiftUI/UIKit (iOS), Android Views (Java/Kotlin), Unity (C#), and PlayStation SDK.
- Hired in part for educational background — role required teaching clients one-on-one how to use the API, strong documentation skills, and ability to develop academic examples and demonstrations.
- Responded to customer issues and feature requests directly, maintaining strong relationships with AAA and Indie studio clients.

`Unreal Engine / C++ / Objective-C / Swift / iOS / Android Native / Java / Kotlin / PlayStation / Nintendo Switch / Xbox / Node.js / Groovy / C# (Unity / Godot) / Shell Script / MacOS / Windows / Linux`

2005 – 2022 **Co-founder / Lead Developer / Instructor** **Silicon Hanna Inc. (self-employed; founded as SuRJE Software Solutions Inc.)**

- Designed, coded, and published eleven games to commercial app stores for mobile devices — including Star Chase and Bitsy (2008, physics-based iOS games released under the SuRJE name) and nine further titles (2011–2022) under the Silicon Hanna brand.
- Developed course curriculum and instructed intermediate C++ programming for the University of Calgary Department of Continuing Education; prepared well-organised lectures, sample code, and assignments; received positive student feedback.
- Technical writing, web development, Google Maps and Firebase integration, and server administration.
- Explored cutting-edge device technologies and SDKs including ARKit augmented reality, Chipmunk physics, and Unreal Engine.

`Unreal Engine / Objective-C / Swift / iOS / Android Native / Kotlin / Augmented Reality / ARKit / C++ / OpenGL / PHP / SQL / MacOS / Windows / Linux`

2008 – 2011 **Tools Programmer** **Fekete Associates Inc. (S&P Global)**

- Developed tools using mathematical models to maximize production and improve efficiency for major oil & gas corporations.
- Contributed to the company's shared UI tools library — the MFC-based component library used as the foundation for all internal development projects, ensuring consistent interaction patterns and visual design across the product suite.
- Led research on a new system that reduced build times by 40%.
- Designed and delivered a voluntary OpenGL elective course for colleagues using the OpenGL Programming Guide as the core reference — covered the rendering pipeline, viewing projections, z-buffering, and geometric transformations; students built an interactive OpenGL widget within the company's custom MFC framework as a capstone project; accommodated both beginner programmers and senior developers; received positive response from students and senior staff.

`C++ / MFC / Windows`

2005 – 2008 **Intermediate Software Engineer** **BJ Pipeline Inspection Services Inc. (Baker Hughes)**

- Visualized in-line inspection device data using OpenGL graphics for Vectra and Gemini 3D analysis products used by pipeline analysts and engineers.
- Implemented neural network algorithms for automated anomaly detection.
- Ported an Excel-based reporting workflow to a full LAMP web application — interactive JavaScript forms, dynamic charts, and interactive geographic maps — dramatically improving accessibility and usability of inspection data for non-technical stakeholders.
- Improved low-level C++ algorithms for high-performance data preprocessing.

`C++ / C / OpenGL / PHP / JavaScript / SQL / Neural Networks / LAMP / Borland / Windows / Linux`

2005 – 2006 **Software Researcher / Lead Developer** **PlayStarMusic Corp.**

- Designed and built the end-user interface for a media streaming storefront — browsing, playback controls, purchase flow, and seamless DRM integration that kept the listening experience uninterrupted.
- Lead developer and system administrator for the full platform stack (iTunes-like storefront, DRM encoding pipeline, video streaming).
- Developed a prototype for a competitive search engine.

`C++ / PHP / Javascript / .NET / Windows / Linux`

2004 – 2005 **Junior Software Engineer** **Codefast.com**

- Primary mandate was building graphical user interfaces for existing command-line tools — translating CLI workflows into usable, discoverable desktop applications.
- Ported applications across GUI libraries (Qt, Wx), ensuring consistent look-and-feel and interaction behaviour under a new UI framework.
- Developed user interfaces for software lifecycle automation tooling.

`C++ / Qt / Wx / Shell script / Windows / Unix`

2002 – 2004 **Web Developer / VR Artist (Contract)** **Nickle Arts Museum, University of Calgary — collaboration with artist Gerald Hushlak**

- Gathered requirements, designed, and implemented the interactive online catalogue for the Gerald Hushlak art exhibit (swarmart.com) — built interactive features using JavaScript.
- Contributed original generative art to the exhibit: Bacterial Chemotaxis visualisations produced from simulation research were displayed alongside the featured works.
- Collaborated with Hushlak on a pioneering VR installation using the University of Calgary CAVE (Cave Automatic Virtual Environment) — 3D-scanned the artist's head, loaded the resulting models into the CAVE, and built an immersive spatial experience years before consumer VR existed.

`C++ / OpenGL / CAVE VR / 3D Scanning / JavaScript / PHP / HTML / CSS`

2001 – 2004 **Research Assistant** **VLab Computer Graphics Group / Artificial Intelligence Research Hub**

- Developed graphics software tools used fractal math and biological modelling.
- Managed network setup and code base for distributed computing.
- Participated in interdepartmental projects and published five research papers.
- Served as teaching assistant and lab instructor across three courses: Introduction to Programming (first year), Computing for Non-Majors (first year), and Operating Systems (third year) — led weekly lab sessions, supplemented instructor lesson plans with independent presentations and code and mathematics walk-throughs, and received positive student feedback.

`C++ / OpenGL / Solaris / MacOS / Unix / Linux / PHP / MySQL / Qt`

EDUCATION

2002–2004 **Master's Degree** **University of Calgary**
Computer Science
Thesis: Agent-based Development of Natural Transportation Networks
Coursework: Human-Computer Interaction, Computer Graphics, Artificial Intelligence

1998–2001 **Bachelor's Degree** **University of Calgary**
Computer Science

SELECTED PUBLICATIONS

- Hoar, R., Penner, J.. "The Application of Artificial Intelligence to Transportation System Design," *XRDS: Crossroads, The ACM Magazine for Students, Vol. 9(3), ACM Press*, 2003. *(Published under maiden name Penner. Ricardo Hoar first author; Joanne second.)*
- Penner, J., Hoar, R., Jacob, C.. "Bacterial Chemotaxis in Silico," *Proceedings of the First Australian Conference on Artificial Life (ACAL), Canberra, Australia*, 2003. *(Best Student Paper Award recipient. Published under maiden name Penner.)*
- Hoar, R., Penner, J., Jacob, C.. "Transcription and Evolution of a Virtual Bacteria Culture," *Proceedings of the IEEE Congress on Evolutionary Computation (CEC), Canberra, Australia*, 2003. *(Published under maiden name Penner.)*
- Penner, J., Hoar, R., Jacob, C.. "Swarm-Based Traffic Simulation with Evolutionary Traffic Light Adaptation," *Proceedings of the IASTED International Conference on Applied Simulation and Modelling (ASM), Crete, Greece*, 2002. *(Published under maiden name Penner.)*
- Hoar, R., Penner, J., Jacob, C.. "Evolutionary Swarm Traffic: If Ant Roads Had Traffic Lights," *Proceedings of the IEEE Congress on Evolutionary Computation (CEC), Honolulu, Hawaii*, 2002. *(Published under maiden name Penner.)*

HONOURS & AWARDS

- Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies, 2005
- Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada, 2005
- Graduate Student Scholarship, Alberta Learning, 2004
- Best Student Paper Award (Co-author), ACAL, 2003
- CPSC Departmental Research Award, University of Calgary, 2003
- Graduate Student Conference Travel Grant, University of Calgary Research Services, 2003
- Academic Travel and Conference Funding, University of Calgary Graduate Students' Association, 2003
- Academic Travel and Conference Funding, University of Calgary Students' Union, 2001

COMMUNITY OUTREACH & LEADERSHIP

- Treasurer, Hanna & District Minor Soccer Association, 2025–present
- Vice President, Hanna Figure Skating Club, 2024–present
- Costume Coordinator, Stage Hanna Junior, 2024
- Women of bitHeads and Cookbook Committee, bitHeads, 2024
- Team Manager/Assistant Coach, U11 Soccer, 2024
- Dryland Training Coordinator, Hanna Figure Skating Club, 2022–2024
- 4-H Public Speaking Judge, District Level Competition, 2022
- Computer Programming for Kids, J.C. Charyk School, 2019–2021
- Reading Program, Our Lady of Fatima School, 2016–2018
- Preschool Gym Leader, YMCA, 2013–2014
- OpenGL Elective Instructor, Fekete Associates, 2010
- Calgary Corporate Challenge, 2007–2008
- Ladies Luncheon Club and Golf Tournament Coordinator, BJ Pipeline Inspection, 2006–2008
- SCIberMENTOR Correspondent / Mentor, 2002–2005
- Vice President, Computer Science Graduate Society, 2002–2003
- Program Committee Member (Peer Reviewer), Agents in Traffic and Transportation Workshop (ATT) at AAMAS 2004, Autonomous Agents and Multiagent Systems, 2004
- VP Academic, Computer Science Undergraduate Society, 2001

LANGUAGES

English — native
French — intermediate
Spanish — rudimentary

WORK SAMPLES

xcodegirl/career: work-samples.

PUBLISHED INTERACTIVE APPLICATIONS

Silicon Hanna Inc. (founded as SuRJE Software Solutions Inc.)

- Ragin' Bull — iOS, Android (2020)
- AR Picture Portal — iOS (2018)
- Fragment Reality — iOS (2018)
- IceCream AR — iOS (2018)
- Jewel Slide — iOS, Android (2017, 2019)
- Word Slide & Find — iOS (2017)
- A Word Theme — iOS, Android (2017, 2019)
- CryptoQuip — iOS, Android (2011, 2019)
- Rafter's Loot — iOS (2011)
- Star Chase — iOS (2008)
- Bitsy — iOS (2008)